

# Integration Digest

$$\int_a^b f'(x) \, dx = f(b) - f(a)$$

## Antidifferentiation

$$\int \left( \frac{df}{dx} \right) dx = f(x) + C$$

$$\int dx = \int 1 \, dx = x + C$$

## TODO

- antidifferentiation patterns are reverse of differentiation
- substitution & integration by parts
- trig integration
- trig subs
  - $c - x^2 \Rightarrow \sin$
  - $c + x^2 \Rightarrow \tan$
  - $x^2 - c \Rightarrow \sec$
- should trig integration even be here since its in trig digest?